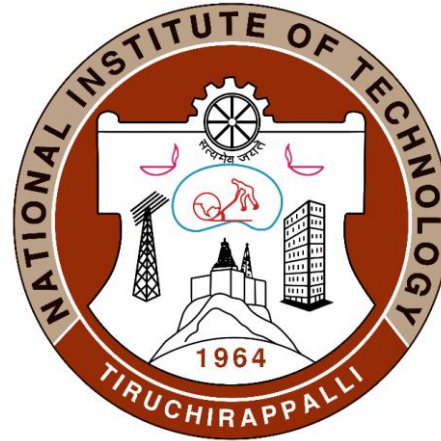


Introduction to Polymers as an Engineering Material



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Lecture - 4

Plastics

Plastics can be classified according to the “level” of performance or the general area of application as shown in the flow diagram

1. High performance Thermoplastics
2. Engineering Thermoplastics
3. Transition Thermoplastics
4. Commodity plastics

The major characteristics of plastics

- They are essentially non-crystalline in nature
- They are non-conductors of heat and electricity
- With some exceptions, they are resistant to chemical and corrosive environment
- They have relatively low softening temperatures
- They exhibit viscoelastic behavior
- They are of low density

Plastics as an engineering materials

- They are inherently low in density and therefore capable of taking more loads
- Many of them such as aramid fibers and UHMWPE posses outstanding specific strength and high specific stiffness
- As a class, they show excellent resistance against most of the corrosive media except some organic solvents and therefore the frequency of replacement needed is less
- Their cost is relatively low when compared with massive metals
- They are generally available in both transparent and translucent varieties

- They show good design flexibility, i.e. they can be molded to any desirable shape
- As a class they are good insulators to electricity but can be made electrically conductive to some extent by embedding metallic filler material. Inherently conductive varieties have also been produced by suitable doping
- They show direct end usability, i.e, they do not need much finishing after being molded or cast
- They can be shaded to a variety of colors
- They are easily available
- Less energy is required for their conversion

Polymeric Materials for Automotive Applications

- ✓ Mechanical components and structures are designed and developed for carrying out certain tasks and are subjected to high stresses, high temperature and corrosive environment.
- ✓ Over the past few decades, high performance plastic materials in automobiles have experienced a real boom and their applications are increasing with a tendency of further growth.
- ✓ The polymers are finding their way deep inside most of the applications primarily because of light weight and relatively better properties.
- ✓ The average global plastic penetration is 120 kg/vehicle and average Indian plastic penetration is 60 kg/vehicle.
- ✓ The basic functions of such wide use of high performance plastic materials in vehicles dictate the appearance of the automobiles, their functionality, economy and low fuel consumption.
- ✓ The application of polymeric materials allows more freedom in design and approximately 82 percent of an average vehicle's Weight gets recycled.

Advantages of increased applications of plastic materials in transport vehicles

- Minimal corrosion, allowing for longer vehicle life
- Substantial design freedom, allowing advanced creativity and innovation
- Flexibility in integrating components
- Safety, comfort and economy
- Recyclability

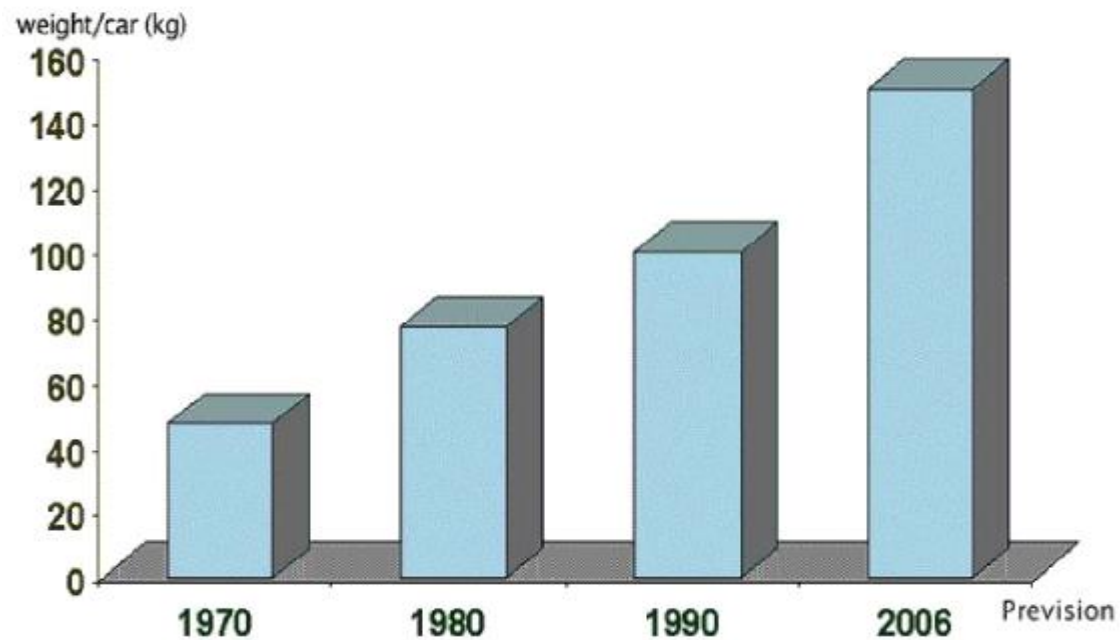


Fig. Increasing use of plastics in automobiles

Table 1. Plastics used in a typical car

Component	Main types of plastics	Average Weight in car (kg)
Bumpers	PS, ABS, PC/PBT	10
Seating	PUR, PP, PVC, ABS, PA	13
Dashboard	PP, ABS, SMA, PPE, PC	7
Fuel systems	HDPE, POM, PA, PP, PBT	6
Body (incl. panels)	PP, PPE, UP	6
Under-bonnet components	PA, PP, PBT	9
Interior trim	PP, ABS, PET, POM, PVC	20
Electrical components	PP, PE, PBT, PA, PVC	7
Exterior trim	ABS, PA, PBT, POM, ASA, PP	4
Lighting	PC, PBT, ABS, PMMA, UP	5
Upholstery	PVC, PUR, PP, PE	8
Liquid reservoirs	PP, PE, PA	1
Total		105

1. PC (Polycarbonate)

They are used mainly in non-reinforced condition. Amorphous polycarbonate polymer offers a rare combination of stiffness, hardness and toughness.

It shows excellent weathering, creep, impact, optical, electrical, thermal properties and UV resistance, with transparency levels almost good as acrylic.

Applications: Bumpers, headlamp lenses, security screens, aircraft panels, spectacle lenses, headlamp lenses, helmets and bullet-proof glass substitutes.

2. Acrylonitrile/butadiene/styrene (ABS)

Durable thermoplastic. It is a copolymer built by polymerizing styrene and acrylonitrile in the presence of polybutadiene.

The styrene gives the plastic a shiny, impervious surface. The butadiene, a rubbery substance, provides resilience even at low temperatures.

A variety of modifications can be made to improve impact resistance, toughness, heat resistance, weather and some chemicals resistance.

Applications: Automotive body parts, dashboards, wheel covers manufacture of housings, Covers and linings.

3.PBT (Polybutylene terephthalate)

Polybutylene terephthalate has good chemical resistance and electrical properties, hard and tough material with water absorption, very good resistance to dynamic stress, thermal and Dimensional stability, They are easy to manufacture due to fast crystallization and fast cooling.

Applications: Fog lamp housings and bezels, sun-roof front parts, locking system housings, door handles, bumpers, carburetor components etc.

4.PET (Polyethylene Terephthalate)

Polyethylene terephthalate has similar conditions as PBT, good thermal stability, good electrical properties, very low water absorption and excellent surface properties. It is mostly used to create synthetic fibres and plastic bottles. You may recognize it on clothing labels under the name “polyester”.

Applications: Wiper arm and their gear housings, headlamp retainer, engine cover, connector housings etc.

5. Polyamide (PA, Nylon 6/6, Nylon 6)

Polyamide is known as nylon 6.6 or nylon 6 and it can be both molded and extruded. Nylon 6/6 has good mechanical properties and wear resistance. They are frequently used when a low cost, high mechanical strength, rigid and stable material is required. They also absorb water easily and components in wet or humid conditions will expand, precluding their use in applications where dimensional stability is required. The main application of polyamide is the manufacture of parts which are under the engine hood, mainly using the types of polyamide (PA) reinforced by fiberglass.

Applications: Gears, bushes, cams, bearings, weatherproof coatings.

6. PVC (Polyvinyl chloride)

This is a material used in automotive industry for the manufacture of the protection for the bottom floor in the car, for internal lining and coating of electric cables in the vehicle. Polyvinyl chloride moulding compounds can be extruded, injection moulded, compression moulded, calendared, and blow moulded to form a huge variety of products, either rigid or flexible depending on the amount and type of plasticizers used. It has good resistance to chemical and solvent attack. Its vinyl content gives it good tensile strength and some grades are flexible. Coloured or clear material is available. PVC has good flexibility, is flame retardant, and has good thermal stability, a high gloss, and low (to no) lead content.

Applications: automobile instruments panels, sheathing of electrical cables, pipes, doors, waterproof clothing, and chemical tanks.

7. POM (Polyoxymethylene)

POM (Polyoxymethylene), also known as Acetal, Polyacetal and Polyformaldehyde, is a high-performance engineering polymer. It is used in precision parts which require high stiffness, low friction and excellent dimensional stability. These properties are stable in low temperatures. POM also is highly chemical and fuel resistant.

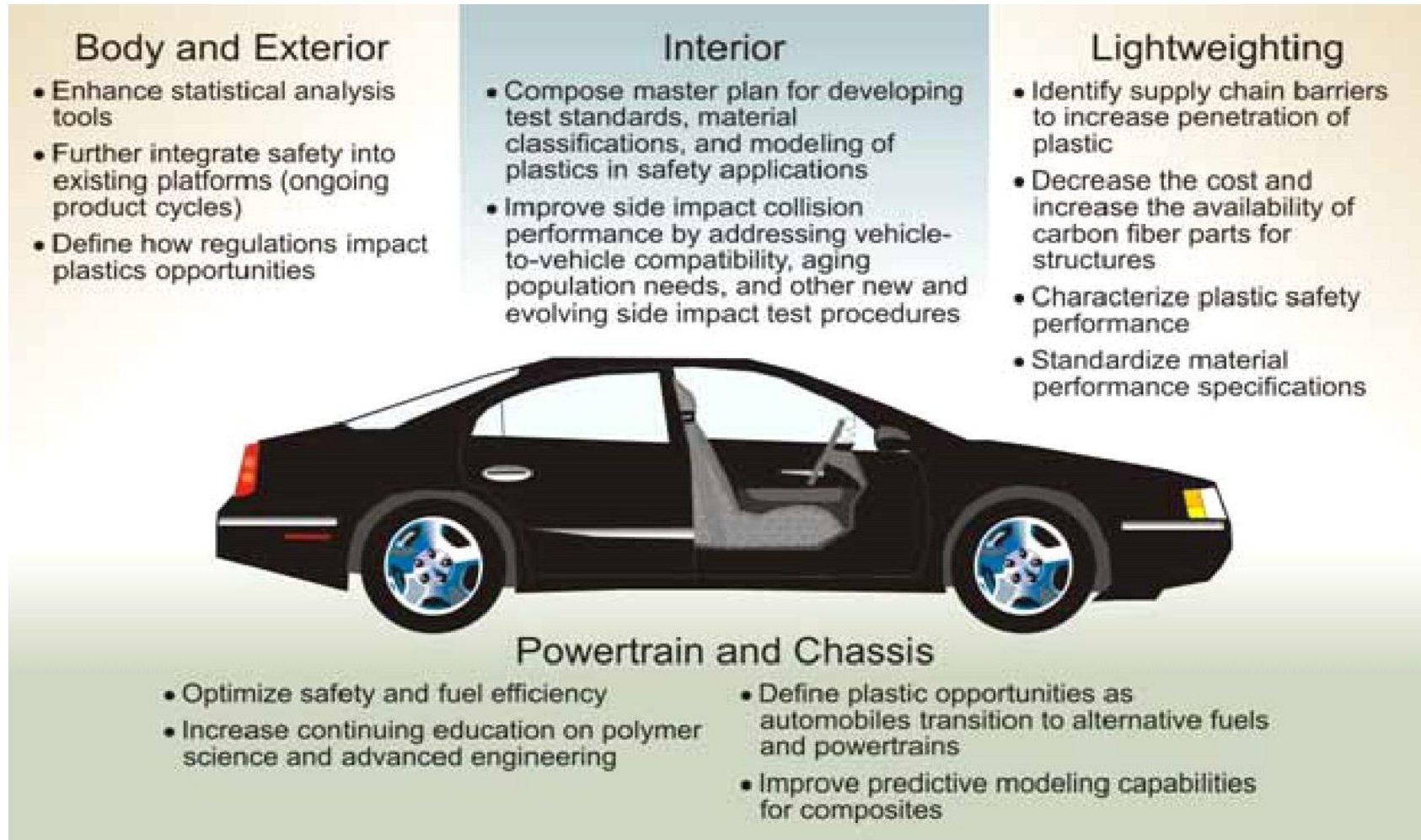
Applications: interior and exterior trims, fuel systems, small gears, window guides, speaker grills, zips, lighters, aerosol valves, fasteners and furniture components.

8. PMMA (Acrylic)

PMMA (Acrylic) is a transparent thermoplastic; it is often used as a lightweight or shatter-resistant alternative to glass and good UV and weather resistance, high optical quality and surface finish with a huge colour range. It's cheaper than PC but is also more prone to scratching and shattering.

Applications: windows, displays, screens.

Highest-Priority Research and Development Needed to Enhance Future Automotive Safety with Plastics



Manufacturing Procedures of Automotive Parts Made of Polymeric Materials

1. *Injection moulding*
2. *Blow moulding*
3. *Heat forming*
4. *Extrusion*

